005930.KS - Samsung Electronics

Full investment research | Model assumptions and source evidence

Owner valuation and shareholder-return update - 28 October 2026

The forward earnings multiples are reduced by 25% from Q4 2026: **3x in Q4 2026; 3.375x in Q1 2027; 3.75x in Q2 2027; 4.125x in Q3 2027; and 4.5x from Q4 2027 onward.** The forecast allocates 50% of positive FCF to shareholder returns, split 41% dividends and 59% buybacks. Ordinary dividends therefore earn **20.5% of positive FCF**, and shareholder buybacks earn **29.5%**. These quarterly allocations are the owner's selected forecast assumptions. Samsung's published 2024-2026 policy returns 50% of cumulative three-year FCF and includes KRW 9.8T in annual regular dividends.

Q4 2027 price ceiling: KRW 497,487.18 per share, versus the previously published KRW 641,449.06. Q1 2028 is KRW 505,569.98. Dividend income appears separately. The operating revenue, product-cost and OP forecast remains intact. Cash-funded buyback cohorts change future economic shares.

Historical forecasts through Q3 2026 retain their published values and then-effective conventions. Q3 2026's preserved price ceiling is KRW 318,581.62. A separate current cash reconstruction bridges the June 30 source balance into Q4.

These return assumptions remain interim until company-specific programmes are announced. Review Samsung's upcoming results for its official return policy, FCF and share count, then update current and future assumptions while preserving published history.

Framework 8.0 | Owner valuation and capital-return update October 28, 2026 | Q3 preliminary guidance dated October 8 | Detailed cash anchor June 30, 2026

HELD. Analytical Base 50/60 = 83/100. Protected portfolio Excel Base 90/100 under the owner override.

Q4 2027: KRW 497,487.18 | Q1 2028: KRW 505,569.98 | Forward-four-quarter operating-profit valuation | Dividend income separately.

The operating model separates HBM, conventional DRAM and NAND. A 50% probability-weighted HBM pricing/mix increment includes production, packaging, yield and operating-cost burdens. The NAND +23%/+20%/+15% calendar path, conventional DRAM residual, non-memory businesses and analytical score remain intact. The capital build begins with 6.648649811B issued common-plus-preferred economic shares and models future cancellation cohorts. Currency is KRW trillions unless stated. Q3 consolidated revenue and OP use preliminary issuer guidance; segment splits remain model estimates. Q4 2026 onward comprises forecast periods.

The HBM source note supplied by the owner was dated October 1 in Korea and incorporated September 30 in Toronto. That historical HBM update is distinct from this October 11 incorporation of the October 8 Q3 preliminary guidance. No future-period forecast is presented as an observed result.

October 11 earnings vintage: Samsung's October 8 preliminary Q3 2026 guidance set revenue at KRW 195T and OP at 107.4T, implying a 55.08% margin. The preceding model had 187.586027357573T revenue and 103.138027357573T OP. Separate unallocated reconciliations preserve the preliminary group total. The October 11 then-6x Q3 ceiling was KRW 477872.43150801185. The October 15 published historical display is KRW 318581.6210053413 at 4x and remains preserved. [SAMS-Q3G]

1. Thesis, products and competitive position

HELD. BOTTLENECK in memory supply, owned through a diversified electronics company. HBM and conventional server memory drive most modeled profit. Foundry, packaging and devices expand technology exposure but also create distinct execution and capital-allocation demands. The owner-selected operating-profit multiple follows the prospective schedule below and reaches 4.5x from Q4 2027.

The bottleneck and the whole company

Samsung combines a large memory manufacturing franchise with foundry/system semiconductor activities, mobile devices, displays and consumer electronics. HBM and advanced memory can possess scarce capacity and qualification advantages while the consolidated business still reflects unrelated product cycles, internal transfers and capital allocation. This is the framework's Rule 10 issue: the attractive niche must be translated into its actual contribution to the whole company.

The approved four-vector operating build separates DS Memory, DS LSI + Foundry, DX net-sales proxy and Other net-sales proxy. HBM, conventional DRAM and NAND are economically distinct within DS Memory, and now have separate modeled revenue, cost and operating-profit schedules. The split is a calibrated proxy; Samsung has not disclosed a complete product-level income statement. HBM requires qualified designs, advanced stacking and packaging yields as well as wafer supply. Conventional DRAM and NAND face different mix and supply cycles. Packaging investment enables the production of chips whose revenue is counted within the Memory pool. Internal memory supplied to another Samsung business cannot be added again to consolidated revenue without the appropriate elimination.

Scale, competition and attribution

Manufacturing scale, process expertise, a broad customer base and financial resources support the user's mega-cap conviction. They do not remove the need to validate HBM qualifications, foundry utilization and the profitability of the non-memory businesses. Samsung, SK hynix and Micron compete on technology, yield, customer qualifications and allocation; a market-wide shortage does not establish a single supplier's product share.

Preferred shareholders participate in economic earnings. The model therefore allocates group operating profit across common and preferred economic claims and reconciles treasury stock. The analytical score remains 83. The current owner-selected portfolio Excel Base is 90. The October 7 override change remains part of the dated history.

Research provenance: the product and competitive analysis in this section synthesizes the supplied original research. Dated market-share, process and channel claims retain their original evidence status; the original verification limits remain attached to those claims. Current earnings and model assumptions are set out below.

Integrated co-design and long-run memory substitution

Samsung's combination of DRAM, logic process, interface engineering and packaging offers a plausible co-design advantage: the supplier can optimize both ends of the memory interface and understand package/process variation earlier. Test that co-design mechanism through accepted output and production cost. A sustained lead depends on qualifications, yield and competitive responses. SK hynix and Micron can also co-design with foundry and interface partners; ownership of every step is not the only route.

Samsung's February 12, 2026 commercial HBM4 announcement specifies 1c DRAM, a 4 nm logic base die and consistent 11.7 Gbps operation, with capability up to 13 Gbps. It predates the article's May 31 assertion that all vendors failed the 11 Gbps requirement. The announcement is company-reported evidence and does not supply a complete yield distribution. The source's SF4X-versus-TSMC and Samsung-versus-other-SiPh rankings lack matched product conditions and commercial-cost data, so they do not create an automatic market-share transfer.

The package measurements reproduced below show why channel characterization matters. They concern the path between an ASIC and HBM's base die; they do **not** measure the internal base-die-to-DRAM stack or disclose Samsung's hidden stack parasitics.

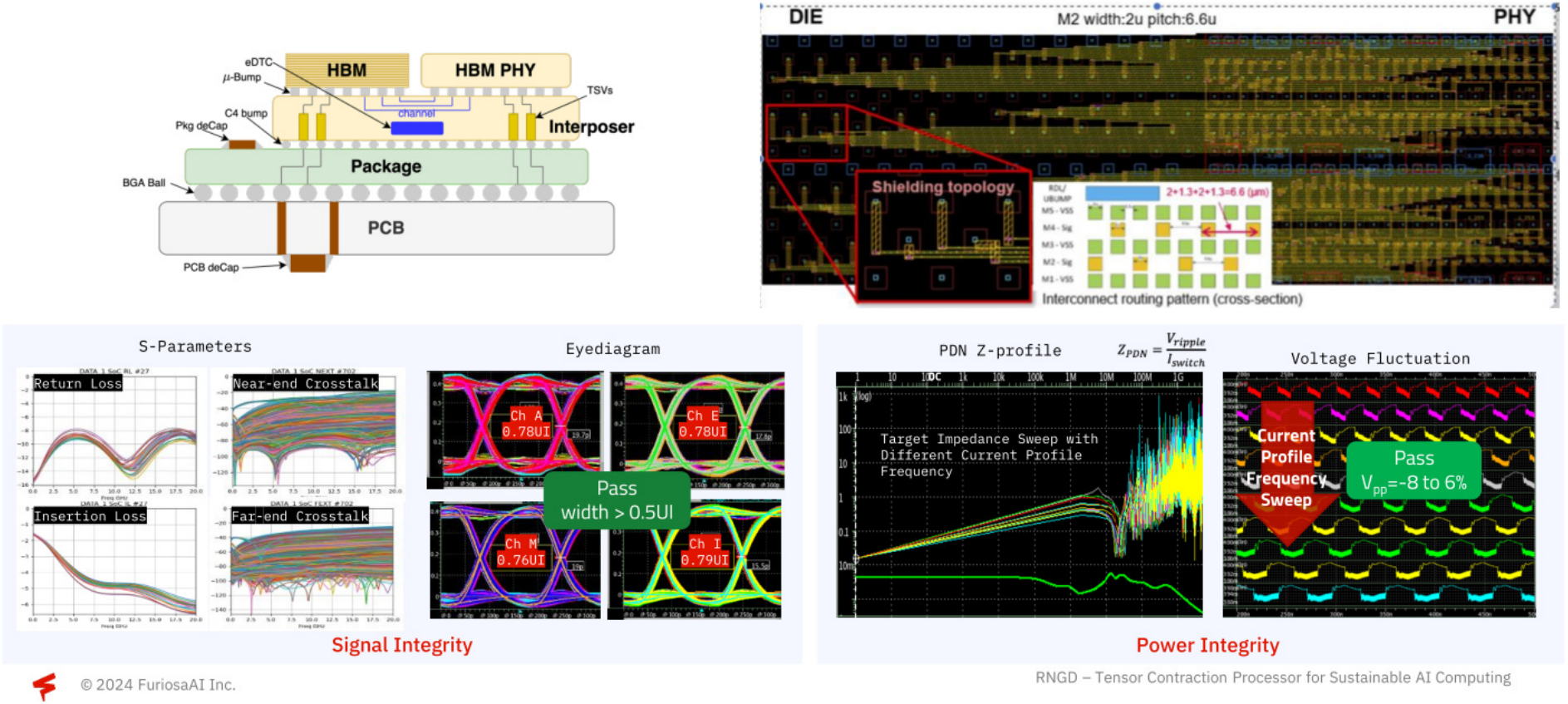


Figure IA24-S1. Furiosa AI Hot Chips 2024 package signal/power-integrity slide reproduced in IA24, body image 113. The exhibit shows the Furiosa package example; Samsung stack measurements and qualification data remain undisclosed.

The existing HBM-01 model already weights the pricing/mix outcome at 50% and includes front-end cost +10%, packaging cost +20%, a 95% to 90% yield stress proxy and additional R&D/SG&A equal to 5% of incremental sales. These are explicit modeled cost and yield stresses. Samsung-specific yield measurements remain undisclosed. The older integration thesis supplies no additional qualified units or independently measurable cost advantage beyond that branch. Internal foundry transfers also do not create another consolidated profit pool. The HBM premium fades in 2029 and disappears in 2030 under the

existing assumption. No extra margin premium, volume shock or Base increase is added.

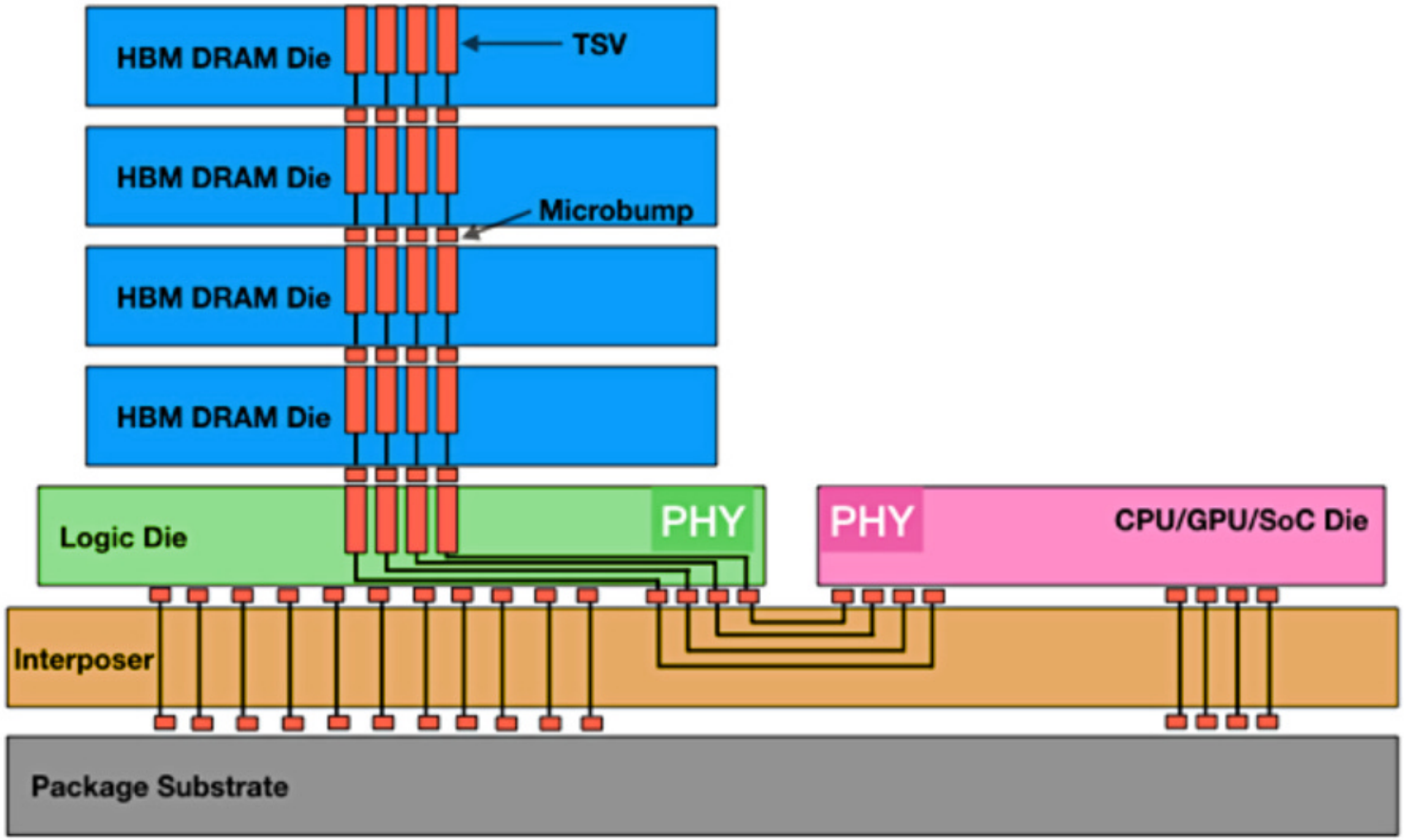


Figure IA24-S2. HBM/base-die/interposer structure reproduced in the May 31 article, body image 5. It maps the production steps relevant to a co-design hypothesis.

Optical memory: promising link physics, incomplete system economics

The May 31, 2026 Irrational Analysis article argues that clock-forwarded optics could eventually connect accelerators to reusable LPDDR pools. HBM is a DRAM product architecture; losing HBM attachment does not imply that its supplier loses every displaced memory bit. Conversely, selling more conventional bits does not guarantee the same revenue or margin. Custom HBM, local HBM caches, LPDDR/SOCAMM and pooled memory can coexist across workloads.

NVIDIA's ISSCC 2026 research demonstrates a link mechanism. Commercial deployment of a disaggregated-memory platform remains a separate milestone. Its supplied comparison table retains thermal-tuning and laser-efficiency assumptions. Link energy and interface density do not measure complete memory-access latency, useful application bandwidth, controller cost or software utilization. The author forecasts NVIDIA timing in 2028-2029 and Broadcom/Marvell timing in 2029-2031; deployment decisions and product execution determine realization. A clock-forwarded link removes some electronics; optical propagation, controller access and resource allocation still take time. CXL pooling has its own protocol and controller architecture. Optical memory pooling also requires qualified links, controllers and software.

Performance Comparison

		This work	VLSI'22 [1]	VLSI'23 [9]	ISSCC'23 [3]
Process	EIC	7nm FinFET	45nm SOI CMOS	28nm CMOS	7nm CMOS
	PIC	65nm SOI SiPh		300mm SOI SiPh	45nm SOI SiPh
Integration		Hybrid	Monolithic	Hybrid	Hybrid
Architecture		FC	EC	EC	EC
Number of wavelengths		9	8	8	7
Per-λ Data rate <Gbps>		32	16	32	50
Per-Fiber Data rate <Gbps >		256	128	256	350
TX+RX Lane size <mm²>		0.013	-	~1	0.031 (RX only)
EIC shoreline density[a] <Tbps/mm>		**0.8**	~0.14	~0.064	-
EIC area density[b] <Tbps/mm² >		**1.33**	~0.064	~0.032	-
RX sensitivity OMA <dBm>		<**-16.0**[c]	-	-10	-10.1 to -11.4
BER		<1E-11	<6.9E-12	<1E-12	<1E-12
Energy efficiency <pJ/b>	Laser	**0.76**[d]	-	-	-
	Thermal	**0.57**[e]|0.49[f]	<0.6	-	-
	TRX + Clock + TT circuits	**1.26**	4.9-5	3.8 (RX only)	0.96 (RX only)
	Total	**2.59**|2.51	-	-	-

[a] **Bi-directional, 1-PHY deep** (0.6mm). [b] Bi-directional. [c] Measured with one data lane and one FWDCLK ON, PRBS7 pattern. [d] Assuming **10%** fiber-coupled laser WPE. [e] Measured with the bench laser. [f] Estimated based on the measured heater efficiency, covering 3σ of ring and laser variation.

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Figure IA24-O. NVIDIA ISSCC 2026 comparison table reproduced in the owner-supplied May 31 article, body image 18. Original research-link metrics and footnotes are preserved; commercial system qualification and unit costs remain separate milestones.

Original table metric	Value / boundary	Model treatment
Data throughput per fiber	256 Gb/s	Link demonstrator; no company shipment assumption

Original table metric	Value / boundary	Model treatment
EIC shoreline density	0.8 Tb/s/mm bidirectional	0.4 in one direction; do not compare to an unlike-direction figure
Total energy	2.59 / 2.51 pJ/bit	Modeled link energy including laser/thermal terms; system memory energy requires additional components
Laser efficiency assumption	10% fiber-coupled wall-plug efficiency	An assumption underlying the energy calculation
Receiver sensitivity condition	One data lane plus one clock lane; PRBS7	Does not establish every-lane, every-workload memory performance

Architecture ledger. The subjective probability of at least one material optical-attached DRAM program by end-2029 is **25%**. The link result and credible engineering path justify nonzero weight; a qualified memory platform's cohort and economics remain distinct. The conditional bridge is displaced HBM sales, plus recaptured LPDDR/other DRAM sales, less avoided HBM wafer/stacking cost, plus replacement DRAM/interface cost and incremental R&D/qualification expense. A bounded proxy was attempted by exposing 0-100% of the company's HBM pool; that range cannot be narrowed to an identifiable accelerator cohort, attachment rate or deployment schedule, and the replacement revenue/cost ratio can change even the sign of the earnings effect. The broad envelope overlaps heterogeneous-memory growth already in the reference model. **The incremental revenue and operating-profit contribution is zero in every current quarter for that specific non-identification and overlap reason**. The source-specific overlap assessment determines the zero increment. The earliest author timing can overlap 2028; this is not dismissed merely as outside the valuation window.

The separate subjective probability of the author's **90% peak-volume HBM decline by 2033-2036 is 20%**. Its conditional endpoint is 10% of a future HBM peak, whose volume unit is not defined by the source. It is outside the forecast ending in 2030 and adds **zero** to the current earnings path. It is the same substitution mechanism as the 25% program hypothesis, so the two probabilities are not multiplied or applied as independent haircuts. This watch hypothesis has zero contribution to the current terminal value or multiple.

Raise the program weight toward 40% when a named architecture provides a defensible affected cohort and comparable memory latency, delivered bandwidth, utilization and total cost; reduce it toward 10% after missed design/qualification milestones. Credible supplier/channel evidence can support that bridge before official commercialization. Revisit the distant 20% outcome only when multiple large architectures show net HBM substitution, including the HBM retained as cache. Mere link demonstrations and reposts do not independently increase either weight.

For Samsung, the replacement bridge must run through the separate HBM and conventional-DRAM vectors: subtract only displaced HBM sales, add only Samsung-captured replacement bits, and reconcile wafer, packaging, foundry and qualification costs on a consolidated basis. It cannot treat freed wafers as immediate saleable bits or stack a new LPDDR market on the existing DRAM residual. With those cohort variables unresolved, the article changes the architecture risks and tests, while the retained operating path and analytical Base 83 remain intact. The current capital and valuation chapters apply the prospective owner multiple schedule, FCF return assumptions and protected Excel Base 90. The October 7 override change remains a dated record.

2. Earnings, guidance and source evidence

October 8 issuer preliminary guidance, incorporated October 11

The issuer guides Q3 2026 consolidated sales of **195T** (range **194-196T**) and operating profit of **107.4T** (range **107.3-107.5T**) under K-IFRS. The headline values are the medians. The resulting 107.4 / 195 = **55.076923076923%** operating margin is our arithmetic on preliminary guidance. Samsung supplies this before external audit is complete; the full earnings call is scheduled for **October 29, 2026, 10:00 KST**. Do not label these figures final actual results. [SAMS-Q3G, SAMS-Q3IR, SAMS-Q3CALL]

Q3 2026 consolidated bridge	Prior model	Preliminary guidance	Unallocated difference
Revenue, KRW T	187.586027357573	195.000000000000	+7.413972642427
Operating profit, KRW T	103.138027357573	107.400000000000	+4.261972642427
Operating costs, KRW T	84.448000000000	87.600000000000	+3.152000000000

The current release does not provide a product or segment bridge. HBM, conventional DRAM, NAND and non-memory forecasts therefore remain explicitly provisional estimates. A separate unallocated revenue residual of **7.413972642427T** and OP residual of **4.261972642427T** reconcile those estimates to the issuer totals. Neither residual is assigned to Memory, product pricing, bit volume or yield. This preliminary earnings release is dated October 8 and covers Q3 2026. TrendForce's October 7 evening EDT relays correspond to October 8 morning KST and refer to this same release.

Rule 16b persistence diagnostic with full overlap treatment. Assign a **25%** subjective probability that part of the unexplained Q3 beat persists for the next two quarters. The conditional range is **0-100%** of the residual per quarter: revenue 0-7.413972642427T, operating cost 0-3.152T and OP 0-4.261972642427T, assuming the same residual mix for this bounded stress test only. The probability-weighted pre-overlap envelope is revenue 0-1.853493160607T, cost 0-0.788T and OP 0-1.065493160607T per quarter; over two quarters OP is 0-2.130986321214T. One persistence outcome covers both quarters; it is not two independent shocks.

Apply a **100% overlap adjustment** to this diagnostic when reconciling it to the active path: Q4 2026 already models **121.08894968382404T OP**, 13.68894968382404T above Q3 guidance, and Q1 2027 already models **144.19721000342682T**. Existing NAND and HBM assumptions already produce substantial further profit growth. The single consolidated beat does not identify additional product pricing, saleable-bit output or a persistent cost reduction beyond that path. The **active incremental revenue, cost and OP contribution from this persistence diagnostic is zero**. That zero follows overlap and lack of a separable incremental driver after the bounded proxy. Stronger separable product economics can change this assessment. Stronger product/mix/cost evidence can change the active model before a final release. Reassess at the October 29 segment disclosure or earlier credible product-specific evidence.

Samsung's July 30 issuer release reports Q2 2026 consolidated revenue of approximately KRW 171.5T and operating profit 89.5T. The official deck separately reports **Memory revenue 120.8T**, DS revenue 127.5T and **DS operating profit 89.2T**. The 89.2T is not standalone Memory profit. DX revenue 48.0T, Display 7.5T and Harman 4.6T require approximately 16.1T of eliminations when combined with DS. The model's non-memory net-sales allocations remain proxies. [SAMS-1, SAMS-2]

The latest company deck describes HBM4 sales growth, HBM4E sampling and server-centered memory demand with continuing capacity constraints. The prior call record gives Q3 DRAM bit growth in the mid-single digits and NAND in the high-single digits. Multi-year agreements and product price floors improve visibility but do not prove an upper price collar or guaranteed future margin for every product. Those July comments support the inherited operating path; the October 8 preliminary earnings release is treated separately above. [SAMS-2, SAMS-3]

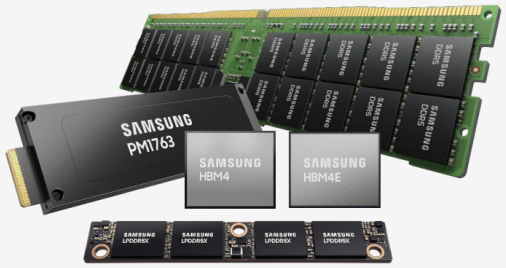
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Memory

Results

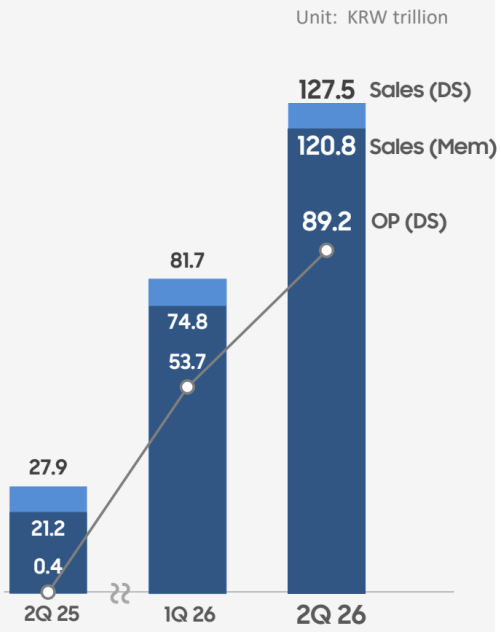
2Q 2026 Results

- **Achieved another record-breaking quarterly earnings**
  - Proactively addressed AI demand with primary focus on server despite limited capacity
  - Continued industry-wide upward price trend
- **Solidified market competitiveness via industry-leading products**
  - Achieved record-high server revenue mix
  - Scaled up HBM4 sales with industry-leading performance
  - Shipped industry's 1st HBM4E samples to major customers



※ Image : Samsung HBM4/E, DDR5 RDIMM, SOCAMM2, V9 eSSD

DS Results



Outlook

H2 2026

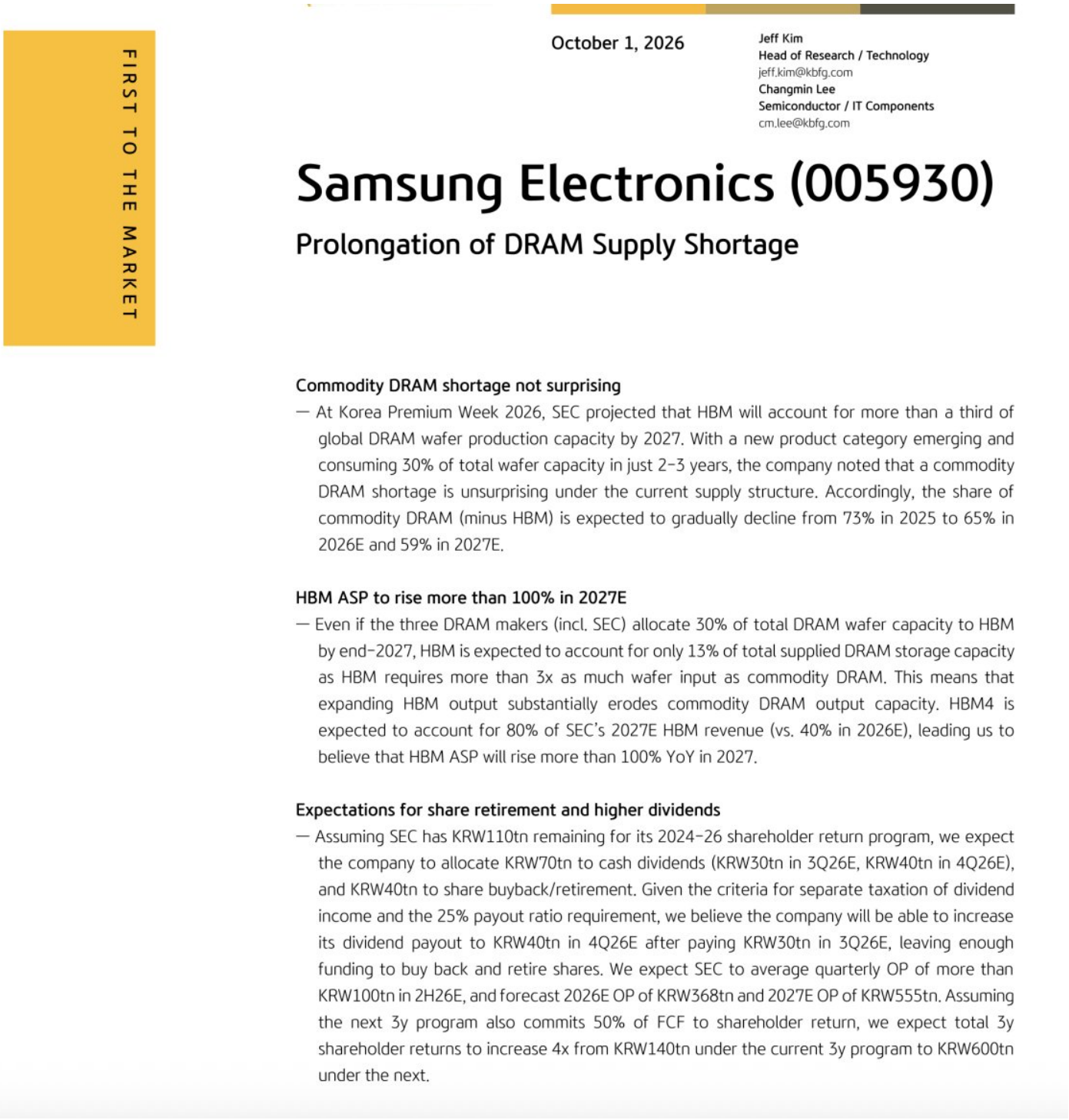
- **Expect robust demand centered on server backed by continued AI Infrastructure CapEx**
  - Strong server demand driven by broader adoption of agentic AI
  - Accelerating growth in demand for server DRAM, Enterprise SSD and HBM expected to keep the market undersupplied despite partial demand moderation in mobile and PC
- **Continue leading the market via technology leadership**
  - Seek to optimize portfolio based on demand trends by application and customer feedback; supply constraints to continue in spite of efforts to increase production
  - Plan to lead the market by addressing strong demand with a focus on high-value-added products, such as HBM4, DDR5, SOCAMM2, etc.
  - Extend the leadership into next-generation AI platforms, following the industry's 1st HBM4E sample shipment as well as the Gen6 and UFS 5.0 market penetration in H1.

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Figure 1. Samsung Q2 2026 earnings presentation, page 7, July 30. Memory sales are reported; HBM qualification and demand commentary support the thesis. DS operating profit must not be misidentified as Memory profit. Source SAMS-2.

New HBM price, mix and supply evidence

The owner-supplied KB Securities page attributes a rising HBM wafer allocation to Samsung's Korea Premium Week presentation. It projects conventional DRAM's share at 73% in 2025,65% in 2026 and 59% in 2027. KB expects HBM4 to represent 80% of Samsung HBM revenue in 2027 versus 40% in 2026, and infers more than 100% HBM ASP growth. These mix and price figures are analyst forecasts for future periods. The original image is preserved here at the point of use. [KB-1]



FIRST TO THE MARKET

October 1, 2026

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Prolongation of DRAM Supply Shortage

Commodity DRAM shortage not surprising

— At Korea Premium Week 2026, SEC projected that HBM will account for more than a third of global DRAM wafer production capacity by 2027. With a new product category emerging and consuming 30% of total wafer capacity in just 2-3 years, the company noted that a commodity DRAM shortage is unsurprising under the current supply structure. Accordingly, the share of commodity DRAM (minus HBM) is expected to gradually decline from 73% in 2025 to 65% in 2026E and 59% in 2027E.

HBM ASP to rise more than 100% in 2027E

— Even if the three DRAM makers (incl. SEC) allocate 30% of total DRAM wafer capacity to HBM by end-2027, HBM is expected to account for only 13% of total supplied DRAM storage capacity as HBM requires more than 3x as much wafer input as commodity DRAM. This means that expanding HBM output substantially erodes commodity DRAM output capacity. HBM4 is expected to account for 80% of SEC's 2027E HBM revenue (vs. 40% in 2026E), leading us to believe that HBM ASP will rise more than 100% YoY in 2027.

Expectations for share retirement and higher dividends

— Assuming SEC has KRW110tn remaining for its 2024-26 shareholder return program, we expect the company to allocate KRW70tn to cash dividends (KRW30tn in 3Q26E, KRW40tn in 4Q26E), and KRW40tn to share buyback/retirement. Given the criteria for separate taxation of dividend income and the 25% payout ratio requirement, we believe the company will be able to increase its dividend payout to KRW40tn in 4Q26E after paying KRW30tn in 3Q26E, leaving enough funding to buy back and retire shares. We expect SEC to average quarterly OP of more than KRW100tn in 2H26E, and forecast 2026E OP of KRW368tn and 2027E OP of KRW555tn. Assuming the next 3y program also commits 50% of FCF to shareholder return, we expect total 3y shareholder returns to increase 4x from KRW140tn under the current 3y program to KRW600tn under the next.

Figure 2. Owner-supplied KB Securities note dated October 1, 2026, Jeff Kim and Changmin Lee. Company-attributed wafer commentary and analyst HBM/earnings/capital-return forecasts have different evidence status. The full exhibit is retained; its analyst valuation and buyback expectations are not adopted.

TrendForce's September 29 original research forecasts **121% growth in blended HBM ASP in 2027**, reflecting constrained output and HBM4/HBM4E mix. Buyers may reduce stack capacity from 12-Hi to 8-Hi, which lowers capacity per stack while increasing base-die cost per bit. The headline ASP denominator is not defined clearly enough to assert it is a pure like-for-like per-bit price. Our use of the forecast at fixed saleable-bit output is an explicit normalization assumption, and contributes to the 50% realization weight. Neither a separate HBM4 premium nor an 8-Hi premium is added. [TF-HBM]

The wafer figures describe directional supply allocation. A Samsung revenue split also requires bit output, prices and product mix. A fall from 65% to 59% is a 9.23% relative decline in allocation only if total wafer capacity is constant. About 10.17% growth in total capacity offsets that allocation decline before productivity and yield changes. The same page's 30%-wafer/13%-bit illustration is not combined mechanically with 59% commodity share as one exact production schedule.

KB projects 2027 company OP 555T, below our inherited 626.710T. This lower forecast provides a counterweight when assessing the inherited operating path. Its capital-return forecasts do not specify a reliable company-authorized cancellation schedule for our share denominator; future cash distributions cannot be added both to the profit valuation and as an independent profit layer.

3. Analytical score

The category arithmetic gives **50/60 = 83/100** after the existing -4-point risk deduction. The current owner-selected portfolio Excel Base is **90/100**. Dated earlier override records remain preserved. Risk-class detail appears in section 7.

Category	Points	Maximum	Basis
Core thesis	17	20	Memory oligopoly and process breadth, assessed at the listed conglomerate level.
Cycle stage	10	10	Current memory allocation and HBM4 ramp.
TAM expansion	10	10	DRAM, HBM, NAND, custom logic and physical AI extend the addressable markets.
Catalysts	5	5	HBM4/HBM4E, advanced foundry nodes and capital-return decisions.
Macro	10	10	AI infrastructure demand and limited qualified capacity.
Alpha	2	5	Local-language contract and share-class work adds detail; broad company coverage limits asymmetry.
Risks	-4	0	Four classes: competitive supply/pricing; HBM/foundry execution; conglomerate governance/capital allocation; customer/device demand and mix.

4. Product decomposition, probability and cost bridge

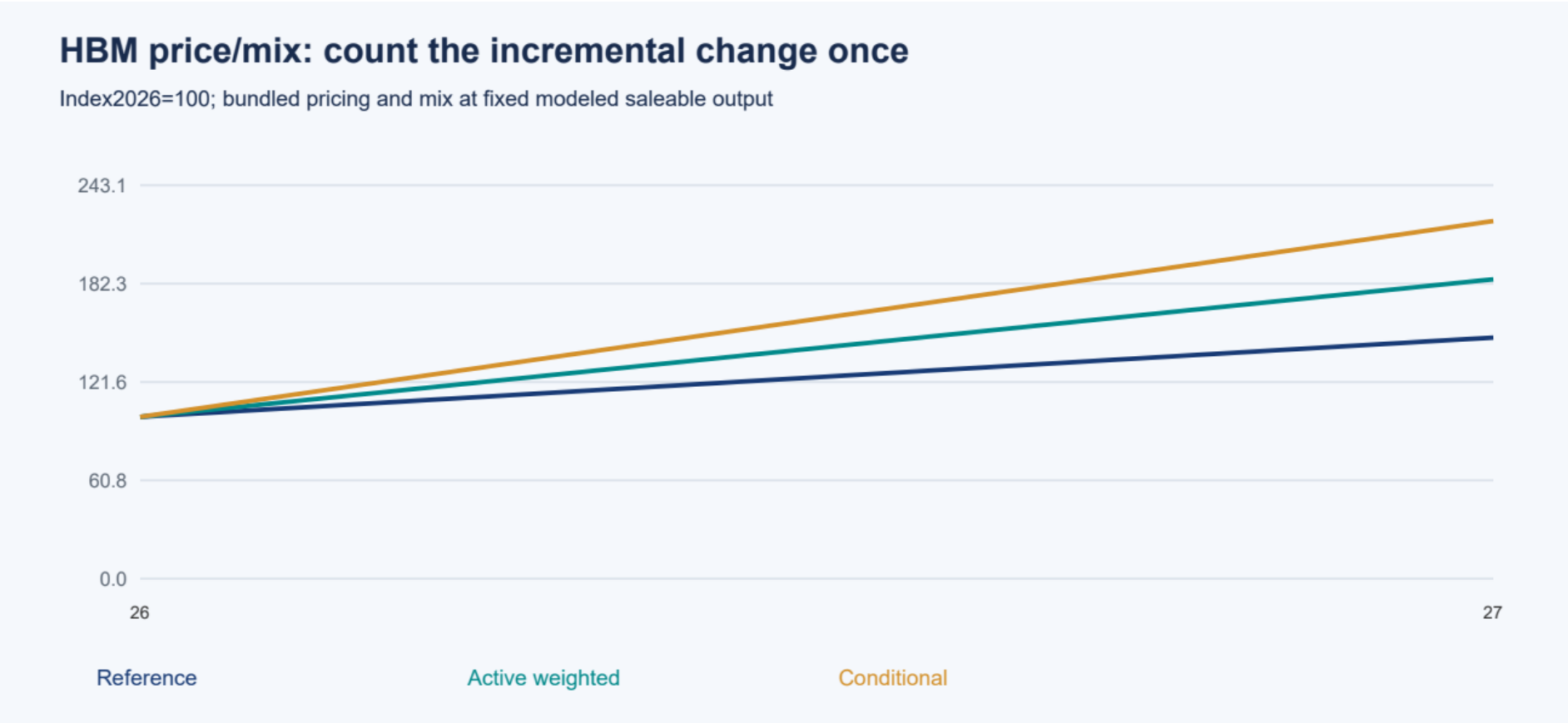
Reconstruct the embedded HBM exposure before repricing

Mirae's May 7 research forecasts 2026 HBM revenue 33.8T. Its September 7 update forecasts 2027 HBM bit growth 57% and ASP growth 49%. These are named-source estimates used to reconstruct an HBM allocation **within** our existing Memory forecast. The old artifact did not explicitly disclose a 49% HBM price assumption; this reconstruction is a reasoned proxy. Historical hidden inputs remain unrecovered. [MA-1, MA-2]

Reference 2027 HBM revenue =33.8 x 1.57 x 1.49 =**79.06834T**. Allocate it across 2027 in proportion to the existing quarterly non-NAND memory revenues, totaling 481.802951T. HBM's reference share is 16.41093% of that pool; conventional DRAM is the residual. For 2026 forward rows, the allocation proxy is 33.8 / (246.9 x 1.443) =9.48701%, using Mirae's 2026 DRAM USD forecast and annual FX assumption. The allocation uses a calendar-average proxy; quarterly product mix can vary. Later reference HBM revenue scales with the retained non-NAND curve at the 2027 allocation. All reference product totals reproduce the original Memory revenue before any new increment.

As a scale cross-check, Mirae's September 2026 DRAM revenue growth and separate HBM/conventional bit-and-price growth imply roughly 34.14T of 2026 HBM revenue, close to the 33.8T anchor. Rounding and FX prevent precision claims. HBM remains a minority of Samsung's memory economics despite its rapid growth. No extra fab output is inserted into the old consolidated trajectory.

Calibration input	Value	Evidence / status
2026 HBM revenue	33.8T	Mirae May 7 forecast; stale but cross-checked scale
Reference 2027 HBM bit growth	57%	Mirae Sep 7 volume forecast
Reference 2027 HBM ASP growth	49%	Mirae Sep 7; reconstructed overlap proxy
Conditional 2027 blended ASP growth	121%	TrendForce industry forecast applied as Samsung proxy
Realization weight	50%	Subjective outcome probability
2027 active blended ASP growth	85%	49% +50% x(121%-49%)
2027 conditional HBM revenue	117.27586T	33.8 x 1.57 x 2.21
2027 active HBM revenue	98.17210T	79.06834 +50% x 38.20752



Modeled incremental HBM price/mix contribution. Numerical table follows.

Index,2026=100	2026	2027
Reference	100	149
Active weighted	100	185
Conditional endpoint	100	221

Explicit probability ledger

HBM-01: 50% probability that Samsung realizes the conditional 121% 2027 blended pricing/mix increase on the calibrated output. TrendForce has the owner's top-tier weighting; KB independently publishes a directionally similar greater-than 100% Samsung estimate, and company ramp evidence supports the product transition. Exact company contracts, headline price units and qualification/stack mix remain uncertain; researcher inputs may share underlying industry checks. The weight applies to realization of the economic outcome. The researcher inputs may share underlying industry checks.

Active HBM revenue =reference HBM revenue +p x(conditional revenue-reference revenue). Active HBM cost =reference cost +p x(conditional cost-reference cost). Both outcomes use the same saleable output, so the calculation adds one pricing/mix increment at fixed saleable output. No increment is assigned to 2026; the 2027 price premium is retained through 2028, halved in 2029, and removed in 2030. The later fade is an uncertainty assumption applied to the conditional premium. Costs fade in the same proportion.

DRAM-01: 60% shortage-persistence judgment, zero separate sales addition. The wafer reallocation signal supports the existing conventional DRAM path; adding another price multiplier from the same shortage would duplicate already modeled scarcity. We explicitly attempted a wafer-share-to-output bridge, but total productive wafers, yield, node productivity and Samsung capture are missing. Allocation alone cannot identify extra sales. This zero reflects overlap and the missing conversion. Stronger separable output economics can change this assessment. The HBM branch still has an active numerical contribution.

Raise HBM-01 toward 65% if company-specific contract pricing and qualification shipments establish comparable-unit conversion; reduce it toward 25% if concessions, lower-capacity stacks or ramp misses dilute the forecast. Revisit the embedded 49% assumption if newer evidence shows more of this price gain was already in the inherited curve. Increase cost stress if base-die, packaging or yield execution disappoints. No independent Base-score increase is awarded for the same profit increment.

Production, packaging, yield and operating expenses

The existing Memory operating-cost envelope is retained as the starting point. Allocate it using revenue-weighted relative cost factors **HBM 2.0, conventional DRAM 1.0, NAND 1.25**, normalized each quarter to sum exactly to the prior cost. These factors are analytical allocation assumptions, reflecting HBM's additional processing and packaging burden; they are not reported product margins. Conventional DRAM's extreme retained scarcity margin remains visible. The split makes that inherited margin risk inspectable.

For HBM,80% of the reference operating cost is allocated to production/COGS and 20% to operating expenses. Production cost is split 65% front-end and 35% packaging. Under the conditional outcome, front-end cost per input rises 10% and packaging cost 20% above the embedded ramp, before a yield adjustment from 95% to 90%. These yield values are modeled stress proxies; Samsung measurements remain undisclosed. Holding saleable output fixed means more input is required; the model does not also subtract units sold. COGS factor =[0.65 x 1.10 +0.35 x 1.20] x 0.95/0.90 =1.19806. Additional R&D/SG&A equals 5% of the incremental HBM sales. Existing reference opex stays in place. These assumptions model mix and ramp costs accompanying the blended-ASP scenario. Saleable output remains fixed within the price comparison.

Conditional HBM COGS =reference HBM cost x 80% x[1 +fade x(1.19806-1)]. Conditional HBM opex =reference cost x 20% +5% x(conditional sales-reference sales). Apply the 50% weight to the resulting total cost difference. Manufacturing cost includes the operating burden of production; capex is not deducted again dollar-for-dollar from operating profit. Company-disclosed gross margins and product opex are unavailable, so these synthetic allocations

must never be presented as reported financials.

Cost assumption	Reference	Conditional
Relative cost weights HBM / DRAM / NAND	2.0 /1.0 /1.25	Reference split preserved
HBM COGS / opex split	80% /20%	Extra opex separately below
Front-end / packaging share of COGS	65% /35%	Same split before cost stresses
Front-end cost per input	1.00	1.10
Packaging cost per input	1.00	1.20
Saleable-yield proxy	95%	90%
Additional R&D/SG&A	None	5% of added HBM sales
Outcome weight	50% reference	50% conditional

2027 HBM economic bridge	KRW T
Reference HBM revenue	79.068
Conditional revenue	117.276
Weighted revenue increment	19.104
Weighted cost increment	2.661
Weighted OP increment	16.443

Existing NAND assumptions

Active NAND pricing and transmission

The active NAND benchmark is **calendar Q3 2026 +23%, Q4 2026 +20% and Q1 2027 +15% sequentially**. Q3/Q4 retain the upper bounds of the owner-supplied TrendForce ranges, ranked Tier 1 alongside Serenity. The added Q1 forecast comes from the owner-supplied Jeff Pu/Jukan summary citing TrendForce. The owner authorized this combined case on September 21. The quoted analyst path was +21%/+17%/+15%; the active case retains the owner's higher 2026 selections. [TF-1, JF-1]

With Q2 2026 indexed to 1.000, active calendar price indexes are **1.230 in Q3, 1.476 in Q4 and 1.6974 in Q1 2027**. The Q1 level is 69.74% above Q2 2026. The +15% is applied once in Q1; it does not recur every quarter. Later forecasts retain their existing operating trajectory at this higher benchmark level. Persistence through 2030 is a modeling assumption selected for this forecast.

The inherited company curves did not contain an auditable NAND ASP/bit split. The current calculation therefore uses **reference-price calibration divisors of 1.150 for Q3 and 1.2075 for Q4 onward**. These reproduce the supplied curve at the previous upper-bound benchmark and serve as reference calibration inputs. Reference-price NAND revenue = calibration NAND revenue / reference index. Active NAND revenue = reference-price NAND revenue x active index. The underlying volume/mix proxy is held constant during this repricing; it is not a measured bit shipment series.

This full-transmission operating case recomputes revenue using one selected price component. The 23% and 20% changes are applied once. This NAND bridge adds no DRAM/HBM price uplift; the separate HBM-01 branch above provides the new HBM increment. The NAND operating bridge holds the existing fab-output assumptions. The separate capital model sets the prospective buyback cohorts. Company contracts, qualification, mix and timing remain the tests of realized earnings.

Samsung NAND allocation and consolidation

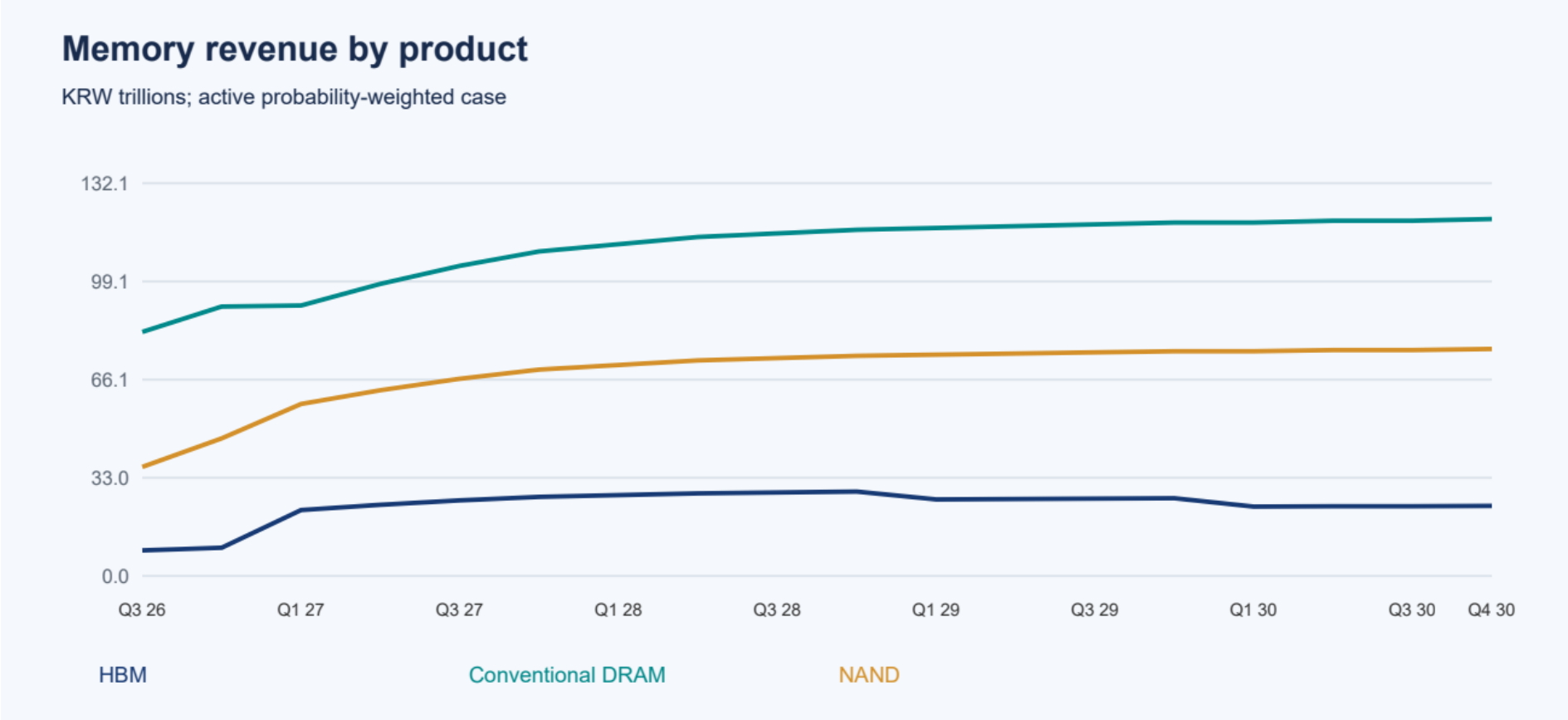
TrendForce's matched Q2 estimates imply NAND / (DRAM + NAND) = **23.06 / (60.98 + 23.06) = 27.4393%**. This ratio allocates the modeled Memory revenue pool; it is not a Samsung-disclosed segment percentage or a new consolidated sales line. Holding that allocation through the forecast is a LOW-confidence modeling assumption. HBM remains within the non-NAND DRAM residual and receives no NAND price uplift. [TF-2, TF-3]

For each quarter, calibration NAND revenue equals the reference Memory curve x 27.4393%. After applying the active price index, Memory revenue equals non-NAND Memory plus active NAND. Reference Memory operating cost, including production expense and divisional opex, is retained in absolute currency from the supplied curve before the separate HBM cost increment. Consequently the incremental price revenue flows one-for-one into Memory OP in this operating ceiling case. Memory OPM is derived from revenue and cost; the HBM branch changes both. NAND and conventional DRAM allocated costs receive no new pricing overlay.

The allocation is applied to the existing consolidated-model Memory pool. No internal sale is added again and DX, foundry and other net-sales proxies remain separate. Existing device-cost pressure stays in the DX path. Any additional demand loss, external NAND purchasing cost, employee incentive expense or unrecovered device margin pressure would reduce group profit relative to this case and is not quantified here. Product scope and internal-elimination differences between industry estimates and company accounts remain an allocation risk.

5. Quarterly operating and valuation model

Calendar quarters are used. Q3 2026 consolidated revenue and OP are issuer preliminary guidance; all Q3 product/segment splits remain LOW-confidence model estimates with a separate unallocated reconciliation. Q4 2026 onward and every product schedule remain modeled. No source provides this full quarterly schedule. Display rounding does not alter exact calculations.



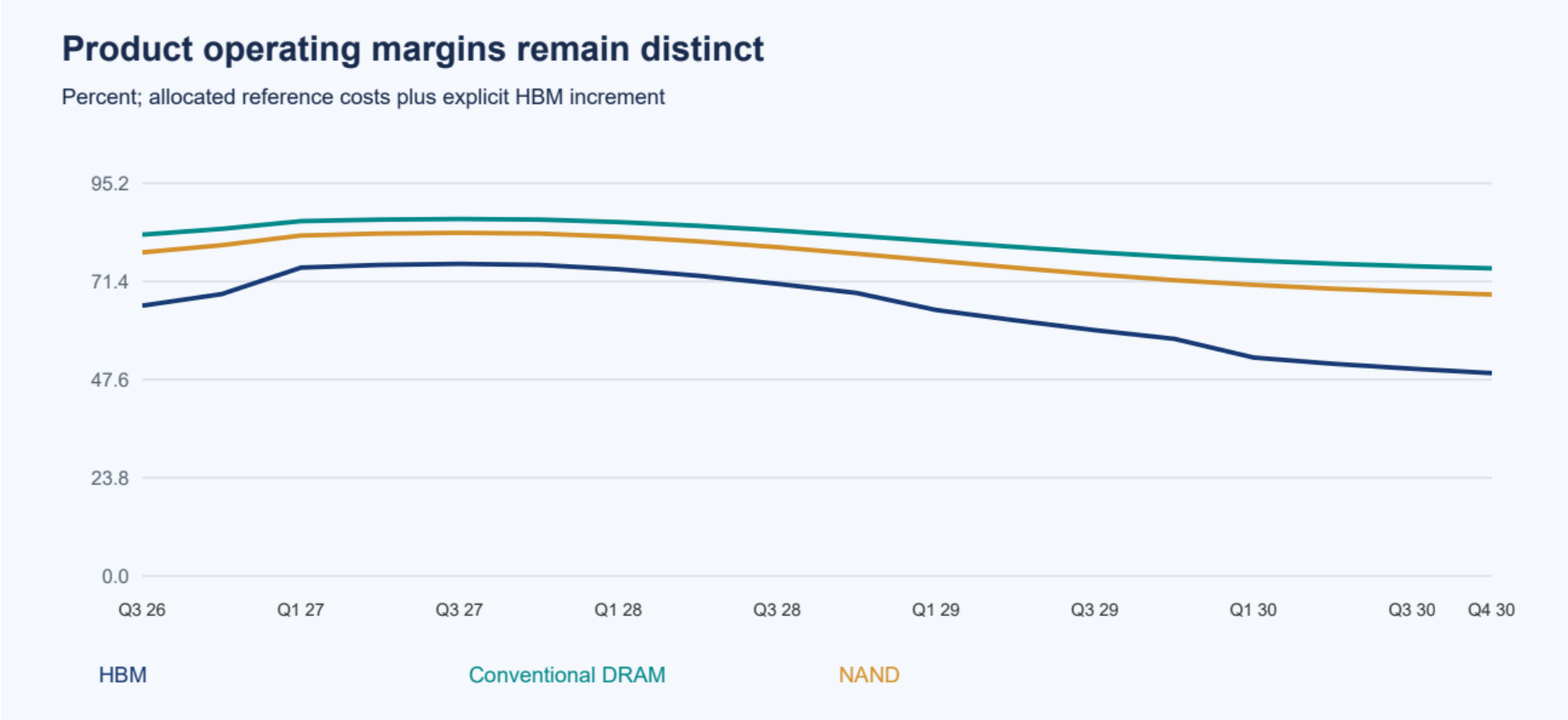
Modeled Memory revenue by product. Numerical table follows.

Quarter	HBM rev	Conv DRAM rev	NAND rev	Memory total
Q3 2026	8.605	82.096	36.685	127.386
Q4 2026	9.500	90.634	46.286	146.420
Q1 2027	22.177	90.979	57.858	171.014
Q2 2027	23.952	98.258	62.486	184.696
Q3 2027	25.430	104.323	66.344	196.097

Quarter	HBM rev	Conv DRAM rev	NAND rev	Memory total
Q4 2027	26.613	109.175	69.429	205.217
Q1 2028	27.204	111.601	70.972	209.778
Q2 2028	27.796	114.027	72.515	214.338
Q3 2028	28.091	115.240	73.286	216.618
Q4 2028	28.387	116.453	74.058	218.898
Q1 2029	25.759	117.060	74.444	217.262
Q2 2029	25.892	117.666	74.829	218.388
Q3 2029	26.026	118.273	75.215	219.514
Q4 2029	26.159	118.879	75.601	220.639
Q1 2030	23.339	118.879	75.601	217.820
Q2 2030	23.459	119.486	75.987	218.931
Q3 2030	23.459	119.486	75.987	218.931
Q4 2030	23.578	120.093	76.372	220.042

Product operating costs and profit

Quarter	HBM op cost	Conv DRAM cost	NAND cost	HBM OP	Conv DRAM OP	NAND OP
Q3 2026	2.964	14.139	7.897	5.641	67.957	28.788
Q4 2026	3.006	14.338	9.153	6.494	76.296	37.133
Q1 2027	5.591	12.685	10.084	16.586	78.294	47.774
Q2 2027	5.882	13.330	10.596	18.070	84.928	51.890
Q3 2027	6.178	13.995	11.125	19.252	90.328	55.218
Q4 2027	6.535	14.811	11.774	20.078	94.364	57.656
Q1 2028	6.966	15.813	12.570	20.239	95.788	58.402
Q2 2028	7.554	17.188	13.663	20.242	96.839	58.852
Q3 2028	8.187	18.674	14.844	19.905	96.567	58.442
Q4 2028	8.905	20.362	16.187	19.482	96.091	57.871
Q1 2029	9.142	22.056	17.533	16.616	95.004	56.910
Q2 2029	9.841	23.767	18.893	16.051	93.900	55.936
Q3 2029	10.511	25.405	20.195	15.515	92.868	55.020
Q4 2029	11.113	26.879	21.367	15.046	92.001	54.234
Q1 2030	10.976	27.954	22.222	12.363	90.925	53.379
Q2 2030	11.386	28.997	23.051	12.073	90.489	52.936
Q3 2030	11.669	29.718	23.623	11.790	89.768	52.363
Q4 2030	11.977	30.502	24.247	11.601	89.590	52.125



Modeled product operating margins. Numerical table follows.

Quarter	HBM OPM	Conv DRAM OPM	NAND OPM	Memory OPM
Q3 2026	65.56%	82.78%	78.47%	80.37%
Q4 2026	68.36%	84.18%	80.23%	81.90%
Q1 2027	74.79%	86.06%	82.57%	83.42%
Q2 2027	75.44%	86.43%	83.04%	83.86%
Q3 2027	75.71%	86.58%	83.23%	84.04%

Quarter	HBM OPM	Conv DRAM OPM	NAND OPM	Memory OPM
Q4 2027	75.44%	86.43%	83.04%	83.86%
Q1 2028	74.40%	85.83%	82.29%	83.15%
Q2 2028	72.82%	84.93%	81.16%	82.08%
Q3 2028	70.86%	83.80%	79.75%	80.75%
Q4 2028	68.63%	82.51%	78.14%	79.24%
Q1 2029	64.51%	81.16%	76.45%	77.57%
Q2 2029	61.99%	79.80%	74.75%	75.96%
Q3 2029	59.61%	78.52%	73.15%	74.44%
Q4 2029	57.52%	77.39%	71.74%	73.10%
Q1 2030	52.97%	76.49%	70.61%	71.93%
Q2 2030	51.46%	75.73%	69.66%	71.03%
Q3 2030	50.26%	75.13%	68.91%	70.31%
Q4 2030	49.20%	74.60%	68.25%	69.68%

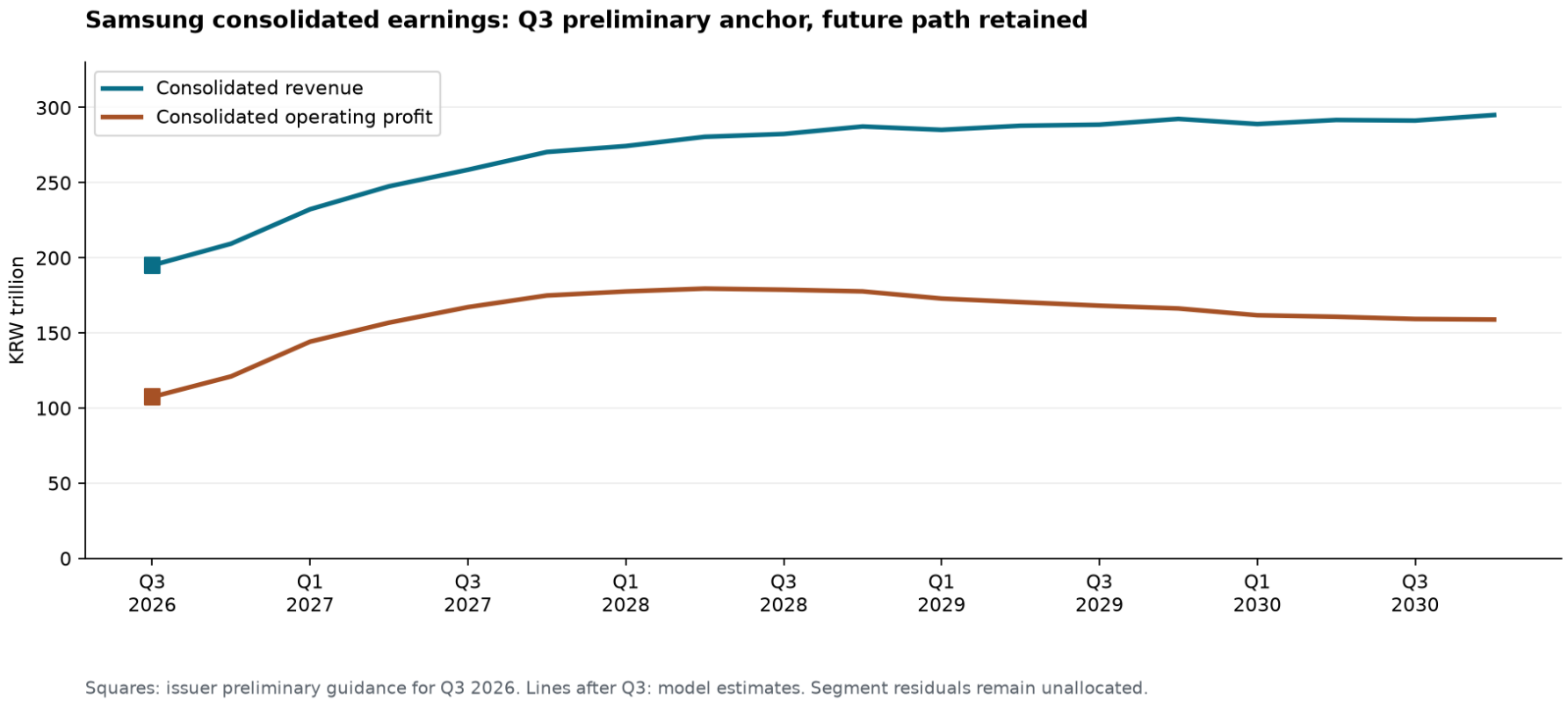
Incremental quarterly bridge

Quarter	Ref HBM rev	Conditional rev	Weighted rev +	Weighted cost +	Weighted OP +
Q3 2026	8.605	8.605	0.000	0.000	0.000
Q4 2026	9.500	9.500	0.000	0.000	0.000
Q1 2027	17.862	26.493	4.316	0.610	3.705
Q2 2027	19.291	28.612	4.661	0.648	4.013
Q3 2027	20.482	30.379	4.949	0.683	4.266
Q4 2027	21.434	31.792	5.179	0.720	4.459
Q1 2028	21.911	32.498	5.294	0.757	4.537
Q2 2028	22.387	33.205	5.409	0.805	4.604
Q3 2028	22.625	33.558	5.466	0.854	4.612
Q4 2028	22.863	33.911	5.524	0.910	4.614
Q1 2029	22.982	28.535	2.776	0.482	2.295
Q2 2029	23.101	28.683	2.791	0.509	2.282
Q3 2029	23.220	28.831	2.805	0.535	2.270
Q4 2029	23.339	28.979	2.820	0.559	2.260
Q1 2030	23.339	23.339	0.000	0.000	0.000
Q2 2030	23.459	23.459	0.000	0.000	0.000
Q3 2030	23.459	23.459	0.000	0.000	-0.000
Q4 2030	23.578	23.578	0.000	0.000	0.000

HBM reference and conditional values are diagnostic components of one weighted operating case. For Q4 2026 onward the prior total-company reference path is recovered when HBM-01 probability is zero; Q3 2026 consolidated preliminary guidance remains fixed and is reconciled separately. No double multiplication by the analytical score or by a final valuation probability occurs.

Retained non-memory businesses

Quarter	Foundry/LSI rev	OPM	DX net rev	OPM	Other net rev	OPM
Q3 2026	16.000	2.00%	38.000	-1.80%	6.200	18.00%
Q4 2026	16.500	2.80%	40.000	-1.20%	6.500	18.20%
Q1 2027	17.000	3.60%	38.000	-0.60%	6.300	18.40%
Q2 2027	17.500	4.40%	39.000	0.00%	6.400	18.60%
Q3 2027	18.000	5.20%	38.000	0.60%	6.500	18.80%
Q4 2027	18.500	5.80%	40.000	1.20%	6.700	19.00%
Q1 2028	19.000	6.40%	39.000	1.80%	6.600	19.10%
Q2 2028	19.500	6.80%	40.000	2.40%	6.700	19.20%
Q3 2028	20.000	7.20%	39.000	2.80%	6.800	19.30%
Q4 2028	20.500	7.50%	41.000	3.20%	7.000	19.40%
Q1 2029	21.000	7.70%	40.000	3.50%	6.900	19.40%
Q2 2029	21.500	7.90%	41.000	3.80%	7.000	19.50%
Q3 2029	22.000	8.00%	40.000	4.00%	7.100	19.50%
Q4 2029	22.500	8.10%	42.000	4.20%	7.300	19.60%
Q1 2030	23.000	8.20%	41.000	4.40%	7.200	19.60%
Q2 2030	23.500	8.30%	42.000	4.50%	7.300	19.70%
Q3 2030	24.000	8.40%	41.000	4.60%	7.400	19.70%
Q4 2030	24.500	8.50%	43.000	4.70%	7.500	19.80%



Consolidated revenue and operating profit. Q3 2026 uses issuer preliminary guidance; Q4 2026 onward comprises forecast periods. Numerical table and unallocated residual follow.

Quarter	Revenue	Memory OP est.	Non-memory OP est.	Unallocated OP	Group OP	Group OPM
Q3 2026 prelim.	195.000	102.386	0.752	4.262	107.400	55.08%
Q4 2026	209.420	119.924	1.165	0.000	121.089	57.82%
Q1 2027	232.314	142.654	1.543	0.000	144.197	62.07%
Q2 2027	247.596	154.888	1.960	0.000	156.848	63.35%

Quarter	Revenue	Memory OP est.	Non-memory OP est.	Unallocated OP	Group OP	Group OPM
Q3 2027	258.597	164.798	2.386	0.000	167.184	64.65%
Q4 2027	270.417	172.098	2.826	0.000	174.924	64.69%
Q1 2028	274.378	174.429	3.179	0.000	177.608	64.73%
Q2 2028	280.538	175.933	3.572	0.000	179.505	63.99%
Q3 2028	282.418	174.914	3.844	0.000	178.758	63.30%
Q4 2028	287.398	173.445	4.208	0.000	177.652	61.81%
Q1 2029	285.162	168.530	4.356	0.000	172.886	60.63%
Q2 2029	287.888	165.887	4.621	0.000	170.508	59.23%
Q3 2029	288.614	163.403	4.744	0.000	168.148	58.26%
Q4 2029	292.439	161.280	5.017	0.000	166.298	56.87%
Q1 2030	289.020	156.668	5.101	0.000	161.769	55.97%
Q2 2030	291.731	155.497	5.279	0.000	160.776	55.11%
Q3 2030	291.331	153.921	5.360	0.000	159.281	54.67%
Q4 2030	295.042	153.316	5.589	0.000	158.905	53.86%

Q3 2026 revenue includes the separate 7.413972642427T unallocated revenue reconciliation. Memory and non-memory OP remain segment estimates; adding the 4.261972642427T unallocated OP reconciles to preliminary group guidance. This residual does not represent a new reporting segment. Future unallocated increments are zero.

Calendar year	Revenue	Operating profit	OPM
2027	1,008.924	643.153	63.75%
2028	1,124.732	713.524	63.44%
2029	1,154.103	677.839	58.73%
2030	1,167.124	640.730	54.90%

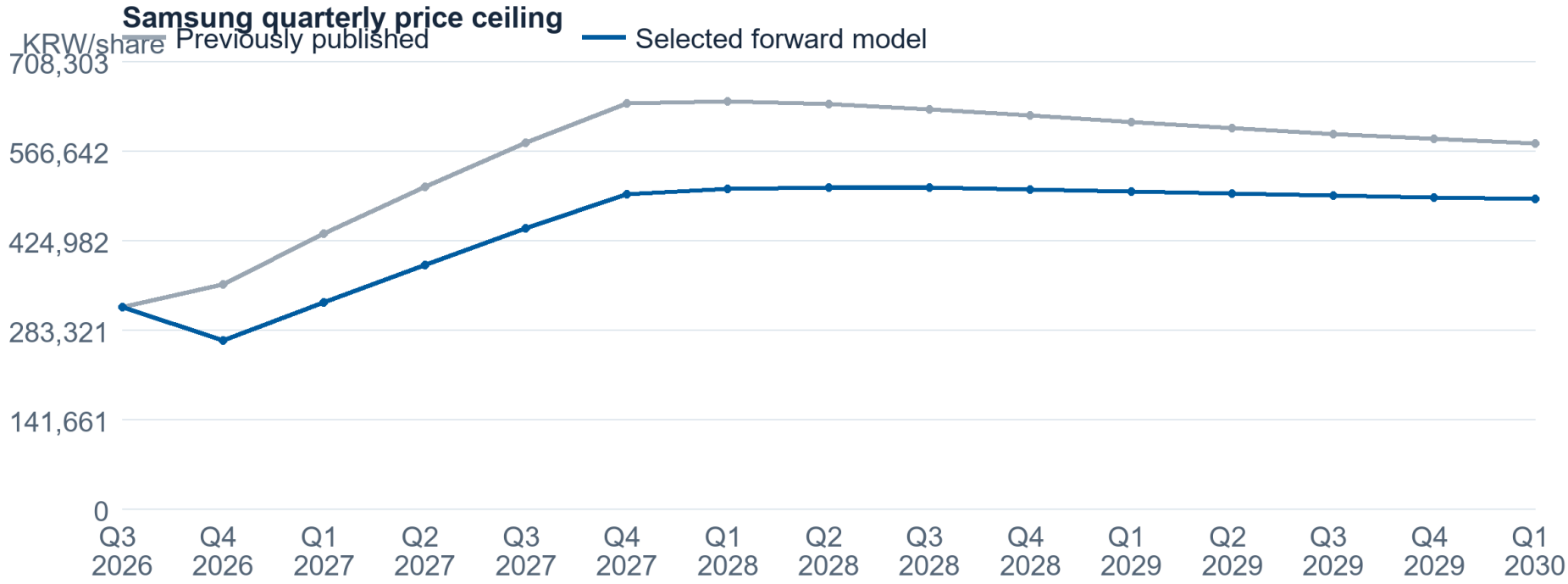
Forward-four-quarter price ceiling

Each future valuation quarter applies its selected multiple to that quarter and the following three quarters of operating profit, divided by modeled end-quarter common-plus-preferred economic shares. The table's dated no-HBM baseline retains the original saved forecast vintage, multiple and share basis, including Q4 2027's KRW 625013.5582978451. Its comparison therefore combines the retained HBM outcome, the owner multiple change and future cancellations. The current price revision versus the immediately preceding published model uses KRW 641449.0626424863 as the Q4 2027 starting point. Dividends are a separate income series. Cash affects payout funding, while valuation uses operating profit and economic shares.

Ceiling quarter	Exact earnings window	F4Q OP T	Multiple	Economic shares B	KRW/share	Change vs dated baseline
Q3 2026	Q3 2026 to Q2 2027	529.534409	4.000x	6.648649811	318,581.62	-32.35%
Q4 2026	Q4 2026 to Q3 2027	589.318127	3.000x	6.648649811	265,911.79	-48.96%
Q1 2027	Q1 2027 to Q4 2027	643.152788	3.375x	6.648649811	326,478.42	-42.27%
Q2 2027	Q2 2027 to Q1 2028	676.563247	3.750x	6.578832965	385,647.76	-35.18%
Q3 2027	Q3 2027 to Q2 2028	699.220324	4.125x	6.507039980	443,255.90	-27.91%
Q4 2027	Q4 2027 to Q3 2028	710.795032	4.500x	6.429467505	497,487.18	-20.40%
Q1 2028	Q1 2028 to Q4 2028	713.523735	4.500x	6.350964106	505,569.98	-19.41%
Q2 2028	Q2 2028 to Q1 2029	708.801934	4.500x	6.276449362	508,186.80	-18.70%
Q3 2028	Q3 2028 to Q2 2029	699.804764	4.500x	6.203611175	507,627.15	-18.00%
Q4 2028	Q4 2028 to Q3 2029	689.194009	4.500x	6.137714534	505,297.70	-17.38%
Q1 2029	Q1 2029 to Q4 2029	677.839230	4.500x	6.076323220	501,993.79	-16.82%
Q2 2029	Q2 2029 to Q1 2030	666.722304	4.500x	6.022109308	498,205.90	-16.34%
Q3 2029	Q3 2029 to Q2 2030	656.989817	4.500x	5.971767397	495,071.89	-15.92%
Q4 2029	Q4 2029 to Q3 2030	648.123013	4.500x	5.928235088	491,976.70	-15.59%
Q1 2030	Q1 2030 to Q4 2030	640.730373	4.500x	5.888690132	489,631.24	-15.32%

- Q3 2026: 529.534408571T x 4.000 x 1,000 / 6.648649811B = KRW 318,581.62. Preserved historical display.
- Q4 2026: 589.318127411T x 3.000 x 1,000 / 6.648649811B = KRW 265,911.79. Selected forward calculation.
- Q1 2027: 643.152787598T x 3.375 x 1,000 / 6.648649811B = KRW 326,478.42. Selected forward calculation.

- Q2 2027: 676.563247174T x 3.750 x 1,000 / 6.578832965B = KRW 385,647.76. Selected forward calculation.
- Q3 2027: 699.220324177T x 4.125 x 1,000 / 6.507039980B = KRW 443,255.90. Selected forward calculation.
- Q4 2027: 710.795031517T x 4.500 x 1,000 / 6.429467505B = KRW 497,487.18. Selected forward calculation.
- Q1 2028: 713.523734503T x 4.500 x 1,000 / 6.350964106B = KRW 505,569.98. Selected forward calculation.
- Q2 2028: 708.801934331T x 4.500 x 1,000 / 6.276449362B = KRW 508,186.80. Selected forward calculation.
- Q3 2028: 699.804763818T x 4.500 x 1,000 / 6.203611175B = KRW 507,627.15. Selected forward calculation.
- Q4 2028: 689.194009461T x 4.500 x 1,000 / 6.137714534B = KRW 505,297.70. Selected forward calculation.
- Q1 2029: 677.839230227T x 4.500 x 1,000 / 6.076323220B = KRW 501,993.79. Selected forward calculation.
- Q2 2029: 666.722303801T x 4.500 x 1,000 / 6.022109308B = KRW 498,205.90. Selected forward calculation.
- Q3 2029: 656.989816624T x 4.500 x 1,000 / 5.971767397B = KRW 495,071.89. Selected forward calculation.
- Q4 2029: 648.123013001T x 4.500 x 1,000 / 5.928235088B = KRW 491,976.70. Selected forward calculation.
- Q1 2030: 640.730372795T x 4.500 x 1,000 / 5.888690132B = KRW 489,631.24. Selected forward calculation.



Samsung price ceilings under the October28 owner selections. Q3 2026 retains its published snapshot.

Sensitivity to uncertain embedded pricing

Each operating case recalculates revenue, operating costs, normalized cash flow, working capital, buyback cohorts and economic shares using the same selected capital law. The HBM probability input is B86, embedded reference ASP is B84 and conditional ASP is B85. These full-precision cases replace the prior rounded diagnostic displays. The selected operating probability remains 50%. The reference allocation remains inside the original total Memory pool; conventional DRAM is the residual, and product costs reconcile to the original Memory cost envelope. The Q3 preliminary-guidance reconciliation stays separate from the HBM branch.

Embedded 2027 HBM ASP

Embedded ASP	Quarter	F4Q OP T	Economic shares B	Price ceiling KRW
49.00%	Q3 2027	699.220324	6.507039980	443,255.90
75.00%	Q3 2027	691.773800	6.506862515	438,547.29
100.00%	Q3 2027	684.622297	6.506690724	434,025.08

HBM outcome probability

Probability	Quarter	F4Q OP T	Economic shares B	Price ceiling KRW
0.00%	Q3 2027	681.354459	6.506180208	431,987.29
25.00%	Q3 2027	690.287391	6.506610094	437,621.96
50.00%	Q3 2027	699.220324	6.507039980	443,255.90
75.00%	Q3 2027	708.153257	6.507469866	448,889.08
100.00%	Q3 2027	717.086190	6.507899753	454,521.53

Share-class and treasury-stock bridge

Samsung reports 5,846,278,608 common and 802,371,203 preferred issued shares at June 30, totaling 6,648,649,811. The 82,086,705 common treasury shares are included in the fully distributed economic basis. H1 cancellations are already reflected in issued shares. Employee purchases and grants are reconciled through that basis. Preferred holders participate in group economics; the model uses equal per-share economic allocation and includes their annual KRW 1 premium within the total dividend pool. [SAMS-4, SAMS-CASH26H1]

The August 21 employee-compensation programme spans August 24-November 21 and approximately KRW 15T. An equal calendar-day spending proxy allocates 38/90 of the programme, or 6.333333T, to Q3 and 52/90, or 8.666667T, to Q4. Actual execution timing remains unverified. Employee purchases receive zero shareholder cancellation credit. The owner forecast purchases common shares under new shareholder-return cohorts, credits 90% cancellation at the end of the second quarter after earning, and spends 70% of each budget in the next quarter and 30% in the following quarter. Execution prices equal the saved KRW 231,000 market reference times 1.10 to the quarter index, with Q4 2026 at index 1. This execution path is independent of the modeled ceilings. [SAMS-5]

The selected case adds zero future issuance beyond the existing fully distributed basis. The 1% annual dilution stress tests additional shares.

Q3 2027 share sensitivity	Economic shares B	Price ceiling KRW
5% fewer	6.181687981	466,585.15
Selected	6.507039980	443,255.90
5% more	6.832391979	422,148.47

Cash flow, dividend income and buyback cohorts

All cash-flow amounts are KRW trillions. The model earns ordinary-dividend budgets of 20.5% of positive quarterly FCF and shareholder-buyback budgets of 29.5%. Q4-earned dividends use a two-quarter payment lag; other earned dividends use a one-quarter lag. The gross dividend income table uses payment-quarter fully distributed economic shares and keeps the annual KRW 1 preferred premium inside the Q4-earned pool. Cumulative income in this table starts in Q4 2026. Source Q2 declared DPS of KRW 374 used eligible shares; the forward distribution proxy uses its stated economic denominator. Unannounced special dividends are zero.

The source balance is June 30 liquid funds of 189.999028T (cash 92.916382T + short-term instruments 97.036646T + short-term FVTPL assets 0.046T) and debt of 22.408721T. The separate Q3 cash reconstruction uses revenue 195T and OP 107.4T, the Q2 declared dividend of 2.455895T and the Q3 employee-programme spending proxy. It generates FCF of 57.331416515T and closing liquid funds of 238.541216182T. The Q4 legacy dividend is a 2.45T proxy. Prior commitments are counted once, alongside the new prospective earning cohorts. [SAMS-CASH26H1]

Normalized FCF = OP + D&A - cash tax - cash interest - change in core trade working capital - gross cash PP&E/intangible purchases - lease principal. H1 revenue of 305.372914T and D&A of 25.374134T give D&A/revenue of 8.3092287615%. H1 cash PP&E purchases of 31.234818T plus intangible purchases of 1.725464T total 32.960282T, giving gross cash capex/revenue of 10.7934530173%. H1 tax expense of 34.418506T divided by pretax profit of 153.268239T gives the normalized 22.4563851092% rate applied to positive OP; source cash tax paid was 6.127297T. June core trade NWC is 150.593405T: receivables 96.408968T + inventory 71.389118T - trade payables 17.204681T. Dividing by Q2 revenue of 171.499470T gives NWC/quarterly revenue of 87.8098369633%. Quarterly cash interest is 0.120367T (H1 0.240734T / 2), and lease principal is 0.3500445T (H1 0.700089T / 2). Future interest and dividend receipts are zero in this conservative funding proxy. Customer financing and asset-sale proceeds receive zero recurring FCF credit. Revenue-linked capital intensity scales with the forecast, including price changes. Cash-tax timing, capital intensity and NWC conversion remain material assumptions.

Selected closing cash stays positive with source debt held as a carry/refinancing proxy. The funding guard rejects negative cash and nonpositive economic common shares. Source June tax payable is 30.303054T; a separate Q4 settlement stress deducts that liability. The source provides no numeric liquidity floor or complete future refinancing schedule. The first cash-distribution column below includes the Q4 legacy dividend payment; the earning columns show new owner-policy budgets.

Quarter	Revenue T	OP T	D&A T	Cash tax T	Change NWC T	Gross cash capex T	FCF T
Q4 2026	209.420	121.089	17.401	27.192	12.662	22.604	75.562

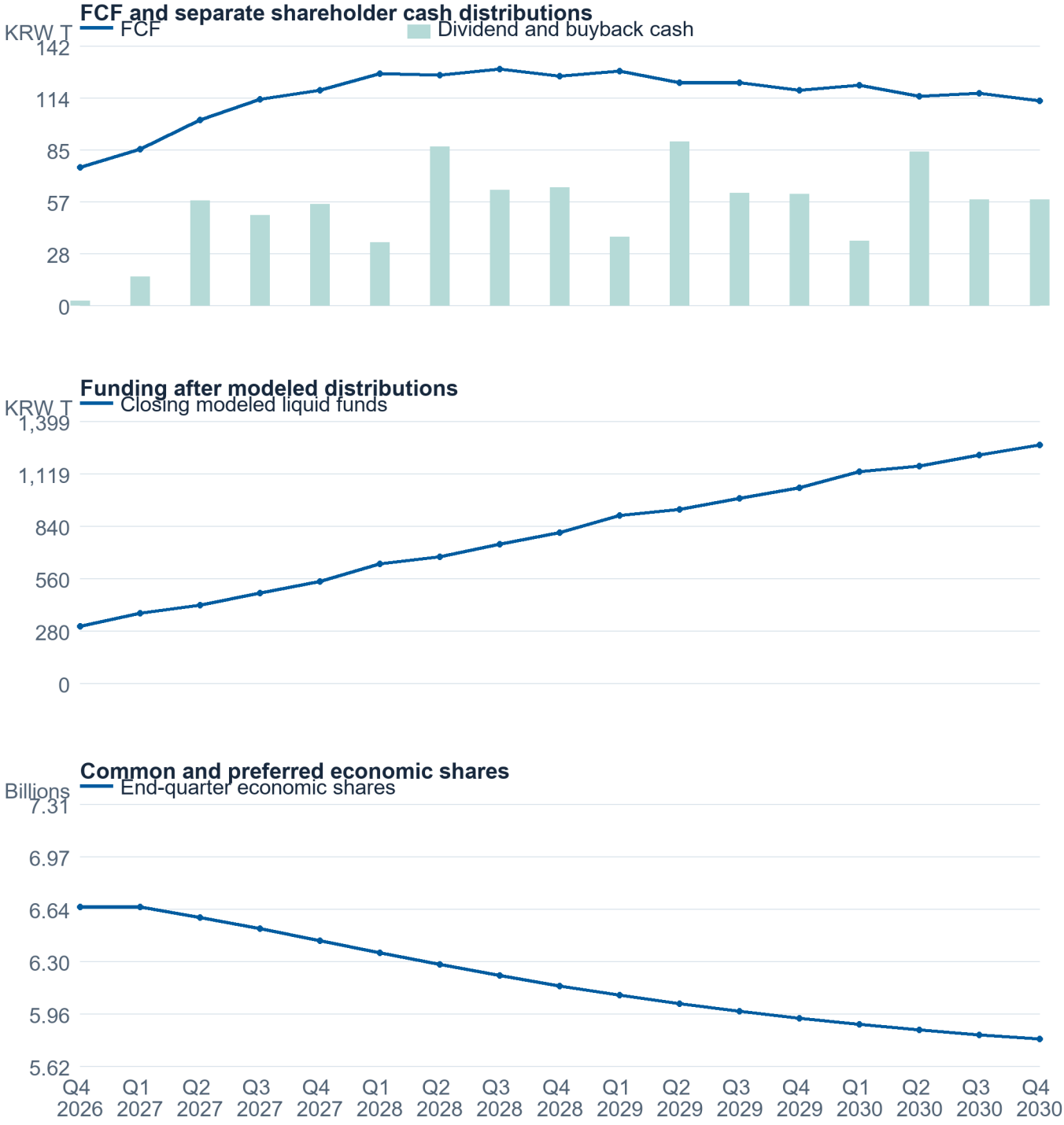
Quarter	Revenue T	OP T	D&A T	Cash tax T	Change NWC T	Gross cash capex T	FCF T
Q1 2027	232.314	144.197	19.304	32.381	20.104	25.075	85.471
Q2 2027	247.596	156.848	20.573	35.222	13.418	26.724	101.586
Q3 2027	258.597	167.184	21.487	37.543	9.660	27.911	113.086
Q4 2027	270.417	174.924	22.470	39.282	10.380	29.187	118.074
Q1 2028	274.378	177.608	22.799	39.884	3.478	29.615	126.959
Q2 2028	280.538	179.505	23.311	40.310	5.409	30.280	126.346
Q3 2028	282.418	178.758	23.467	40.143	1.651	30.483	129.478
Q4 2028	287.398	177.652	23.881	39.894	4.373	31.020	125.775
Q1 2029	285.162	172.886	23.695	38.824	-1.964	30.779	128.471
Q2 2029	287.888	170.508	23.921	38.290	2.393	31.073	122.203
Q3 2029	288.614	168.148	23.982	37.760	0.637	31.151	122.110
Q4 2029	292.439	166.298	24.299	37.344	3.359	31.564	117.859
Q1 2030	289.020	161.769	24.015	36.327	-3.003	31.195	120.794
Q2 2030	291.731	160.776	24.241	36.104	2.381	31.488	114.573
Q3 2030	291.331	159.281	24.207	35.769	-0.351	31.445	116.156
Q4 2030	295.042	158.905	24.516	35.684	3.259	31.845	112.162

Quarter	Dividend earned T	Buyback earned T	Dividend paid T	Buyback cash T	Employee cash T	Economic shares B	Liquid funds T
Q4 2026	15.490	22.291	2.450	0.000	8.667	6.648649811	302.986
Q1 2027	17.521	25.214	0.000	15.603	0.000	6.648649811	372.853
Q2 2027	20.825	29.968	33.012	24.337	0.000	6.578832965	417.091
Q3 2027	23.183	33.360	20.825	28.542	0.000	6.507039980	480.810
Q4 2027	24.205	34.832	23.183	32.343	0.000	6.429467505	543.359

Quarter	Dividend earned T	Buyback earned T	Dividend paid T	Buyback cash T	Employee cash T	Economic shares B	Liquid funds T
Q1 2028	26.027	37.453	0.000	34.390	0.000	6.350964106	635.928
Q2 2028	25.901	37.272	50.232	36.667	0.000	6.276449362	675.375
Q3 2028	26.543	38.196	25.901	37.326	0.000	6.203611175	741.626
Q4 2028	25.784	37.104	26.543	37.919	0.000	6.137714534	802.939
Q1 2029	26.337	37.899	0.000	37.431	0.000	6.076323220	893.979
Q2 2029	25.052	36.050	52.120	37.660	0.000	6.022109308	926.401
Q3 2029	25.033	36.023	25.052	36.605	0.000	5.971767397	986.855
Q4 2029	24.161	34.768	25.033	36.031	0.000	5.928235088	1,043.650
Q1 2030	24.763	35.634	0.000	35.145	0.000	5.888690132	1,129.300
Q2 2030	23.487	33.799	48.924	35.374	0.000	5.853991929	1,159.574
Q3 2030	23.812	34.266	23.487	34.350	0.000	5.821662486	1,217.893
Q4 2030	22.993	33.088	23.812	34.126	0.000	5.793785734	1,272.117

Quarter	Common DPS paid KRW	Preferred DPS paid KRW	Cumulative common DPS KRW
Q4 2026	368.50	368.50	368.50
Q1 2027	0.00	0.00	368.50
Q2 2027	5,017.73	5,018.73	5,386.23
Q3 2027	3,200.41	3,200.41	8,586.63
Q4 2027	3,605.68	3,605.68	12,192.31
Q1 2028	0.00	0.00	12,192.31
Q2 2028	8,003.10	8,004.10	20,195.41
Q3 2028	4,175.13	4,175.13	24,370.54
Q4 2028	4,324.59	4,324.59	28,695.13

Quarter	Common DPS paid KRW	Preferred DPS paid KRW	Cumulative common DPS KRW
Q1 2029	0.00	0.00	28,695.13
Q2 2029	8,654.71	8,655.71	37,349.84
Q3 2029	4,194.99	4,194.99	41,544.84
Q4 2029	4,222.61	4,222.61	45,767.45
Q1 2030	0.00	0.00	45,767.45
Q2 2030	8,357.19	8,358.19	54,124.64
Q3 2030	4,034.49	4,034.49	58,159.13
Q4 2030	4,109.91	4,109.91	62,269.03



Modeled Samsung FCF, separate cash distributions, liquid funds and economic shares. Complete tables follow the selected cash law.

Capital-return and funding sensitivities

Each case recomputes the selected Q4 2027 price, share and funding outputs. The 1% annual dilution case adds economic shares prospectively. The tax-liability settlement case deducts the June liability in Q4, alongside normalized current taxes. Payout, execution-price, cancellation, capex and working-capital stresses expose the dependence of the per-share result on cash conversion and capital allocation. The selected case retains 50% payout and the 41%/59% split. Legal cancellation and dividend eligibility may differ from the modeled fully distributed proxy.

Case	Quarter	Price ceiling KRW	Economic shares B	Liquid funds T
No new shareholder-return cohorts	Q4 2027	481,086.80	6.648649811	721.203
Selected FCF policy with previous multiples	Q4 2027	663,316.24	6.429467505	543.359
All payout cash to dividends	Q4 2027	481,086.80	6.648649811	533.351
All payout cash to buybacks	Q4 2027	509,558.54	6.277154378	550.314
100% positive-FCF payout	Q4 2027	515,045.21	6.210285200	365.515
0% cancellation	Q4 2027	481,086.80	6.648649811	543.359
100% cancellation	Q4 2027	499,378.73	6.405113916	543.359
Half execution-price path	Q4 2027	515,045.21	6.210285200	543.359
Double execution-price path	Q4 2027	489,149.56	6.539058658	543.359
Double cash capex intensity	Q4 2027	492,708.80	6.491821591	439.497
25% higher NWC intensity	Q4 2027	496,737.14	6.439175588	490.762
Cash tax 10 percentage points higher	Q4 2027	494,772.43	6.464745080	484.181
1% annual economic dilution	Q4 2027	491,170.59	6.512152225	543.359
June tax liability settled in Q4	Q4 2027	497,487.18	6.429467505	513.056

6. Industry relationships and read-throughs

TAM expansion and cross-book

AI inference expands conventional DRAM, HBM and enterprise storage together. Custom base dies, packaging, device AI and robotics provide adjacent demand; they must not be counted both as internal input sales and external system sales. LPK's glass-processing opportunity is a possible equipment flow-through whose Samsung order capture remains unconfirmed.

Cross-watch SK hynix and Micron on HBM qualification, supply growth and conventional memory allocation; watch SanDisk/Kioxia for NAND demand and contract terms. Samsung's price-floor disclosure supports a more stable contracted business but does not independently prove a floor for every future quarterly gross margin.

Samsung memory demand can validate industry investment in packaging and tools, but it does not establish an LPKF order or another supplier's capture. Strong memory economics may pressure Dell, Longsys and Penguin through replacement costs. Compare physical bit output and product mix before transferring a revenue-growth inference.

The active NAND benchmark is shared with the SNDK and Kioxia updates, but issuer exposure, recognition dates and operating cost treatment differ. The same percentage increase in NAND prices does not imply the same percentage increase in consolidated earnings or per-share value.

7. Risks and falsification tests

The following ledger makes the existing aggregate risk deduction explicit. It does not introduce another deduction or change the analytical score.

Risk class	Existing deduction
Competitive supply, pricing and allocation	-1
HBM qualification, packaging yield and foundry execution	-1
Conglomerate governance and capital allocation	-1
Customer, device demand and consolidated mix	-1
Total	**-4**

Risks and triggers

1. Supply/pricing response: test the shortage-duration thesis against qualified new capacity and actual pricing and accepted output as new factories ramp.
2. HBM/foundry execution: monitor HBM4 conversion, HBM4E qualification and Taylor production acceptance. Customer-specific process wins determine revenue timing.

3. Governance/capital allocation: distinguish compensation purchases from cancellation, dividends and investment. Track preferred/common claims and future grants.
4. Demand/mix: monitor premium device sales, component inflation and server-versus-consumer mix.

The next earnings update should replace the HBM allocation, price normalization and synthetic product costs with stronger product/bit/ASP evidence and reconcile the share bridge to the latest dilution note. The operating ceiling remains conditional on the inherited memory path.

Numbered monitoring tests

#	Checkpoint	Evidence supporting the operating case	Counter-test / failure signal
1	HBM progress	Track qualified shipments and yield to revise the existing probability and capture assumptions; credible pre-qualification signals remain modelable under Rule 16b.	Qualification delays or disappointing yield reduce expected capture and timing; they do not impose an official-confirmation gate on breadcrumbs.
2	Consolidation	Reconcile memory and other businesses to group revenue and OP.	Unexplained eliminations or duplicate internal sales invalidate the bridge.
3	Economic shares	Reconcile common, preferred, treasury and dilutive instruments to the latest notes.	A common-only denominator overattributes group economics.
4	Diversification	Track profit conversion in foundry, devices and the residual businesses.	Memory strength offset by non-memory losses weakens consolidated leverage.

NAND pricing tests

Checkpoint	Required observation	Failure signal
Realized ASP	Track calendar Q3 +23%, Q4 +20% and Q1 2027 +15% benchmark against company NAND mix and realized ASP.	Contract caps, timing or discounts cause weaker transmission.
Output and cost	Reconcile saleable bits, mix, costs and incentive expense to the fixed reference-price curve.	Price gains are absorbed by lower units or additional costs.
Persistence	Test whether Q1 price gains persist as assumed in later forecasts.	ASP cuts or new capacity erase the retained price premium.

Additional tests for this update: separate a true per-bit price gain from larger stacks or richer products; confirm saleable-bit output and factory yield without counting the same volume twice; test the 49% embedded-price proxy against the inherited curve; monitor process, packaging and incentive costs. The inherited high memory profitability remains the largest uncertainty. The near 12-16 months carry more useful forecasting weight than 2029-2030; the explicit premium fade reflects that deterioration in visibility.

8. Sources, evidence gaps and model snapshot

- **SAMS-Q3G** - October 8, 2026, Samsung issuer Q3 preliminary guidance. Consolidated K-IFRS revenue 195T / OP 107.4T; ranges 194-196T /107.3-107.5T. Issuer preliminary guidance ahead of the final release. Original HTML retained with SHA-256.
- **SAMS-Q3IR** - October 8, 2026, Samsung IR pre-earnings disclosure 84775. Confirms units and preliminary status before external audit completion.
- **SAMS-Q3CALL** - Retrieved October 11, 2026, Samsung IR events. Full Q3 call scheduled October 29 at 10:00 KST.

Sources below are those documented in the supplied earnings review. LOW-confidence product allocations, future margins, capacity bridges and incomplete management-call records remain unresolved where stated. The current revision applies Framework 8.0 probability and visual requirements. Inherited sources are distinguished from the new evidence; their original gaps are not represented as newly resolved.

- **SAMS-1** · 2026-07-30 · Korean company Q2 results. Actual operating results and outlook, original language.
- **SAMS-2** · 2026-07-30 · Company Q2 presentation. Divisional revenue includes internal sales; consolidation requires eliminations.
- **SAMS-3** · 2026-07-30 company event · Q2 management transcript. Management speech: HBM, bit growth, LTAs and foundry capacity. Host displays July 29; company dates results July 30.
- **SAMS-4** · 2026-06-30 share date · Company listing information. Common, preferred and treasury share reconciliation.
- **SAMS-5** · 2026-08-21 · Issuer RNS: share repurchase decision. Employee compensation purpose and planned purchase window; the cited evidence uses the issuer RNS text.
- **SAMS-6** · 2026-09-01 · Uploaded model and artifact - User upload: v3_2_master_portfolio(1).xlsx; 005930KS_Samsung_DeepDive.md. This September 1 source records the then-sanctioned 6x multiple and operating vector path.

- **SAMS-CASH26H1** - Samsung 2026 half-year business report, source period June 30, 2026: financial statements pp42-52, borrowings p76, capital/dividends p79, cash-flow adjustments p84, liquidity pp85-86 and shareholder-return policy pp147-148. The linked model retains source dates and its separate current reconstruction.

Original source files are retained in the external version archive.

- **TF-1** - Owner-supplied TrendForce revision, incorporated September 20, 2026: Q3 NAND ASP +18-23% and Q4 +15-20% sequentially. Active selections: +23% and +20%. The exact revised publication/table was not independently retrieved. Source priority is Tier 1 alongside Serenity as instructed by the owner.
- **TF-2** - August 18, 2026, TrendForce NAND supplier research. Samsung NAND revenue estimate $23.06B; SK hynix Group includes Solidigm. The article's aggregate and individual supplier totals do not reconcile; its aggregate and market-share denominator are not used in the model.
- **TF-3** - September 7, 2026, TrendForce DRAM supplier research. Samsung Q2 DRAM revenue estimate $60.98B, including the DRAM product mix. The Samsung allocation uses matched Q2 product estimates in USD, avoiding a new FX conversion.
- **JF-1** - Supplied September 21, 2026: Jeff Pu (@sssjeffpu) and Jukan (@jukan05), "NAND Pricing Breakout Driven by eSSD" summary in the owner's conversation. Quoted Q3 2026 / Q4 2026 / Q1 2027 blended NAND changes: +21% / +17% / +15%, citing TrendForce demand research. Only the Q1 +15% is newly adopted; the owner retains +23% and +20% for 2026. This is an owner-authorized combined case. The original analyst note and exact TrendForce table were not independently retrieved. Enterprise demand commentary is contextual and adds no shipment volume to this update.
- **KB-1** - Owner-supplied image of KB Securities, Jeff Kim / Changmin Lee, dated October 1, 2026: HBM wafer allocation, HBM4 revenue mix, greater-than 100% HBM ASP hypothesis and capital returns. Original image embedded in section 2. Original report URL not supplied; attribution comes from the visible page.
- **TF-HBM** - TrendForce, September 29, 2026: 121% blended HBM ASP forecast,8-Hi/12-Hi tradeoffs and supply constraints. Industry forecast; Samsung contract realization remains uncertain.
- **MA-1** - Mirae Asset, May 7, 2026, page 1: 2026 HBM revenue 33.8T forecast. This dated sizing anchor is used only for calibration; its older 2027 price estimate is superseded by MA-2.
- **MA-2** - Mirae Asset, September 7, 2026, pages 1 and 5: 2027 HBM bits+57%, ASP+49%,2026 DRAM USD 246.9B and annual FX 1,443. These reconstruct embedded exposure; the analyst's target and total-company forecast are not imported.
- **SAMS-2E** - Samsung Q2 English earnings deck, July 30, 2026, page 7: Memory revenue 120.8T and HBM/server-demand outlook. Source image embedded in section 2.

Model ownership and remaining gaps: probability 50%, relative cost weights, COGS/opex split, front/packaging cost stresses, yield proxies, price-unit normalization and 2028-2030 persistence are explicitly analytical. No measured Samsung HBM yield or standalone HBM operating margin has been invented. Reconcile new issuer evidence against these controls when available. Source reputation raises the weight of a credible forward signal; it does not turn the future outcome into certainty.

Current model binding: Q4 2027 price KRW 497487.17745677714; F4Q OP 710.7950315173491T; multiple 4.5x; economic shares 6.429467505433286B. Analytical Base 83; protected Excel Base 90. History through Q3 2026 remains preserved. Canonical SHA-256: 0ef90ba3185585f09bbe870b50a830c59e45ff6d8b77de3f92d22fca3667254d.